

The TIMEID Procedure

Input Data Set	
Name	WORK.PLAYSTORE
Label	
Time ID Variable	Date
Time Interval	MONTH

Output shows time ID summary, duplicate dates, and gaps in the monthly sequence

The TIMEID Procedure

Time Component				
Value Index	Date	Offset	Span	Interval Count
1	JAN2012	20	.	1
2	FEB2012	19	1	1
3	MAR2012	21	1	1
4	APR2012	20	1	2
5	MAY2012	12	1	1
6	JUN2012	20	1	1
7	JUL2012	19	1	1
8	AUG2012	20	1	1
9	SEP2012	21	1	1
10	OCT2012	14	1	1
11	NOV2012	15	1	1
12	DEC2012	20	1	2
13	JAN2013	19	1	1
14	FEB2013	18	1	1
15	MAR2013	18	1	1
16	MAY2013	17	2	1
17	JUN2013	16	1	1
18	JUL2013	17	1	1
19	OCT2013	16	3	1
20	NOV2013	17	1	1
21	DEC2013	16	1	1
22	JAN2014	0	1	1
23	FEB2014	1	1	1
24	MAR2014	2	1	1
25	APR2014	3	1	1
26	MAY2014	4	1	1
27	JUN2014	5	1	1
28	AUG2014	6	2	1
29	SEP2014	10	1	1
30	OCT2014	30	1	1
31	NOV2014	0	1	1
32	DEC2014	0	1	1

Output shows time ID summary, duplicate dates, and gaps in the monthly sequence

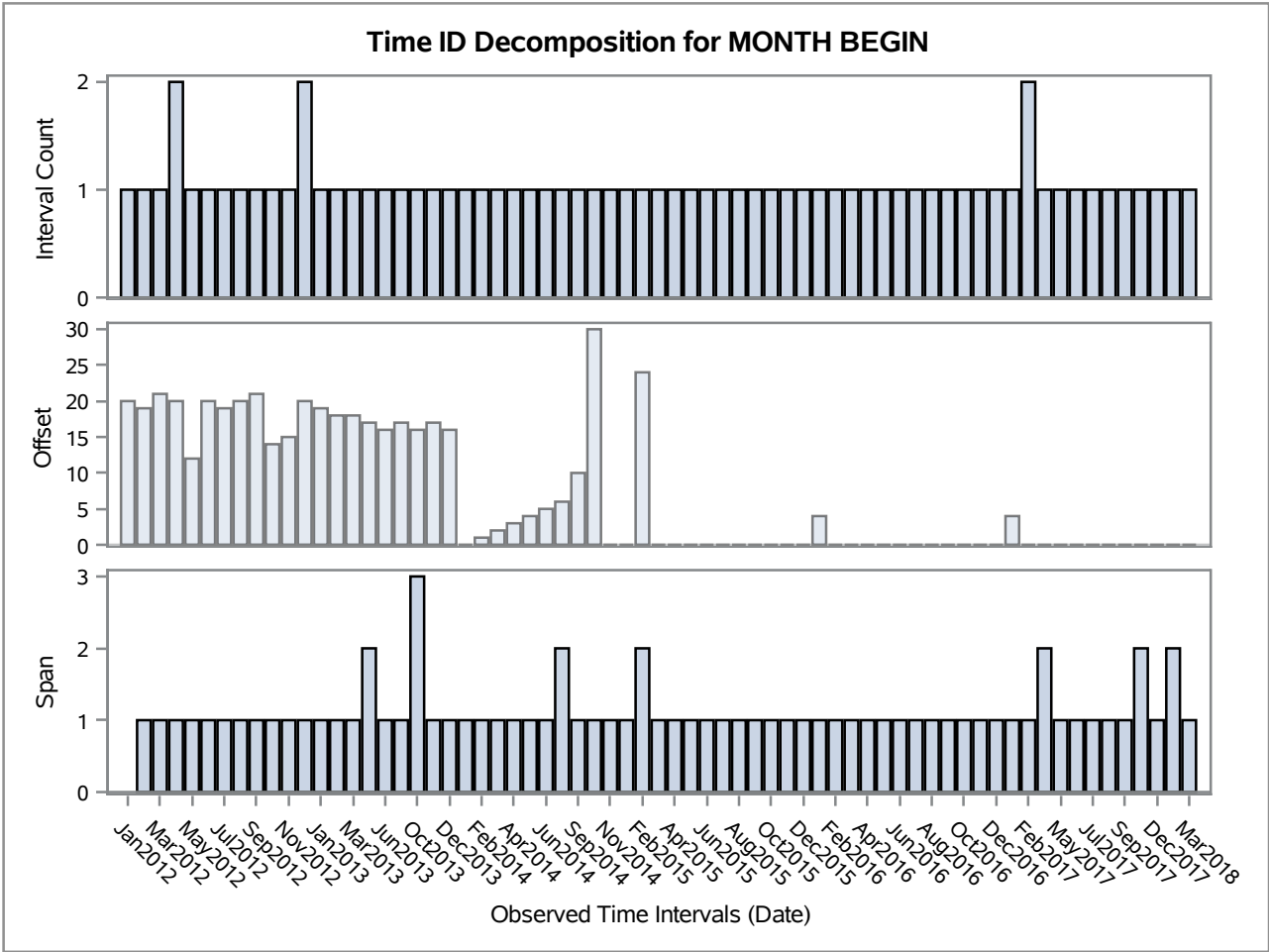
The TIMEID Procedure

Time Component				
Value Index	Date	Offset	Span	Interval Count
33	FEB2015	24	2	1
34	MAR2015	0	1	1
35	APR2015	0	1	1
36	MAY2015	0	1	1
37	JUN2015	0	1	1
38	JUL2015	0	1	1
39	AUG2015	0	1	1
40	SEP2015	0	1	1
41	OCT2015	0	1	1
42	NOV2015	0	1	1
43	DEC2015	0	1	1
44	JAN2016	4	1	1
45	FEB2016	0	1	1
46	MAR2016	0	1	1
47	APR2016	0	1	1
48	MAY2016	0	1	1
49	JUN2016	0	1	1
50	JUL2016	0	1	1
51	AUG2016	0	1	1
52	SEP2016	0	1	1
53	OCT2016	0	1	1
54	NOV2016	0	1	1
55	DEC2016	0	1	1
56	JAN2017	4	1	1
57	FEB2017	0	1	2
58	APR2017	0	2	1
59	MAY2017	0	1	1
60	JUN2017	0	1	1
61	JUL2017	0	1	1
62	AUG2017	0	1	1
63	SEP2017	0	1	1
64	NOV2017	0	2	1

Output shows time ID summary, duplicate dates, and gaps in the monthly sequence

The TIMEID Procedure

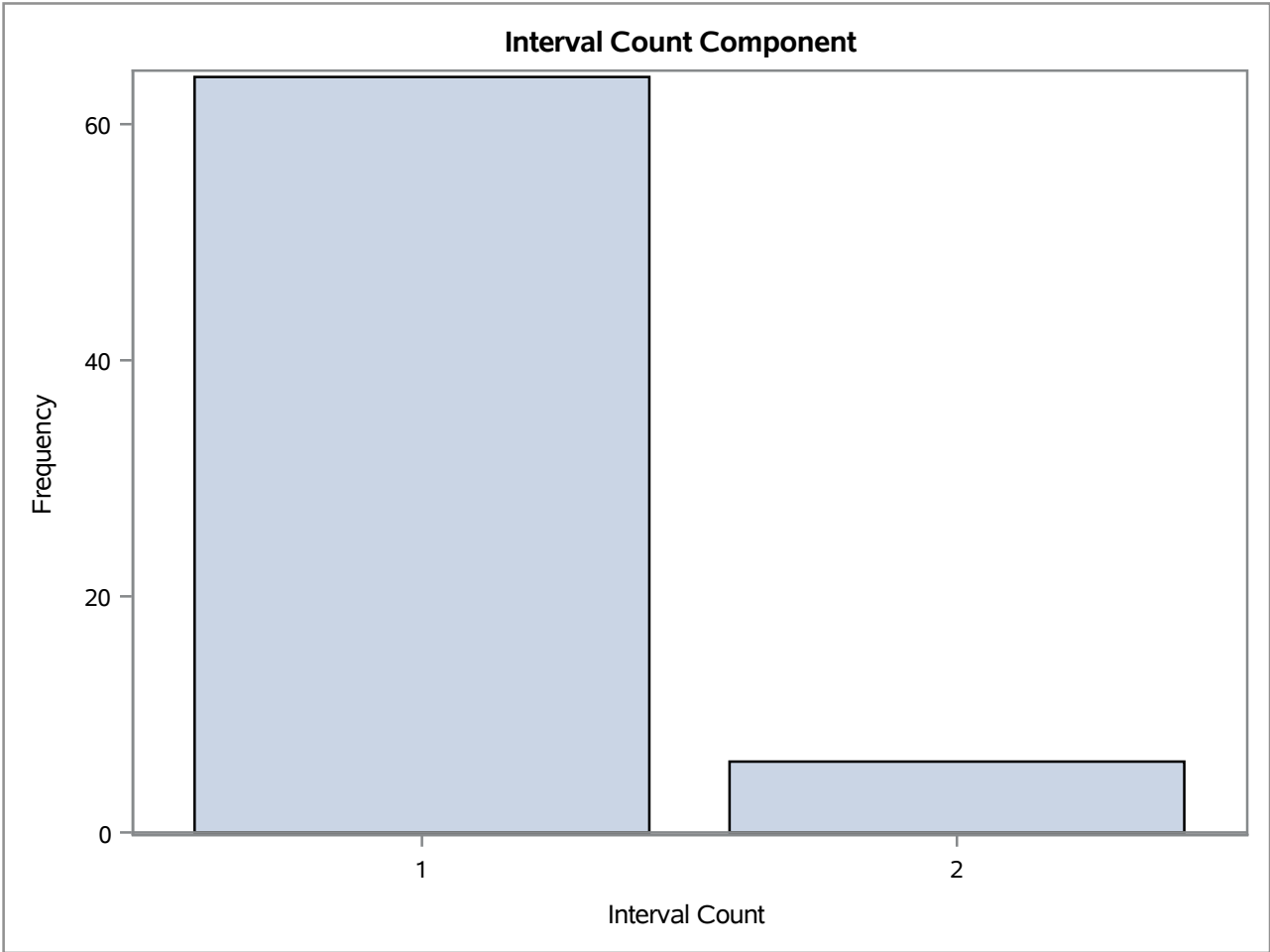
Time Component				
Value Index	Date	Offset	Span	Interval Count
65	DEC2017	0	1	1
66	FEB2018	0	2	1
67	MAR2018	0	1	1



Output shows time ID summary, duplicate dates, and gaps in the monthly sequence

The TIMEID Procedure

Component			
Value Index	Interval Count	Frequency	Percentage
1	1	64	91.428571
2	2	6	8.571429



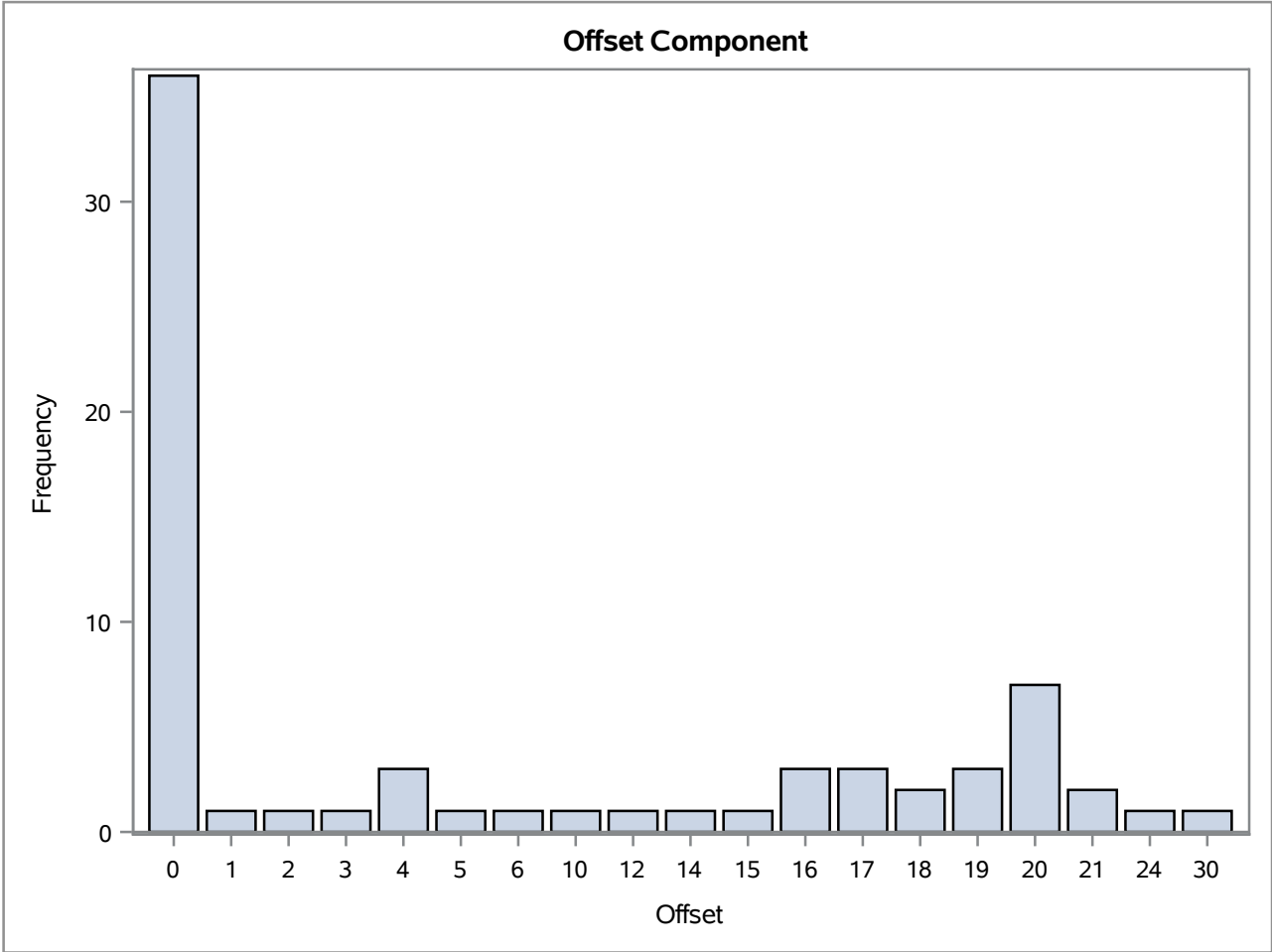
Statistics Summary			
Minimum	Maximum	Mean	Standard Deviation
1	2	1.0857143	0.93395

The TIMEID Procedure

Component			
Value Index	Offset	Frequency	Percentage
1	0	36	51.428571
2	1	1	1.428571
3	2	1	1.428571
4	3	1	1.428571
5	4	3	4.285714
6	5	1	1.428571
7	6	1	1.428571
8	10	1	1.428571
9	12	1	1.428571
10	14	1	1.428571
11	15	1	1.428571
12	16	3	4.285714
13	17	3	4.285714
14	18	2	2.857143
15	19	3	4.285714
16	20	7	10
17	21	2	2.857143
18	24	1	1.428571
19	30	1	1.428571

Output shows time ID summary, duplicate dates, and gaps in the monthly sequence

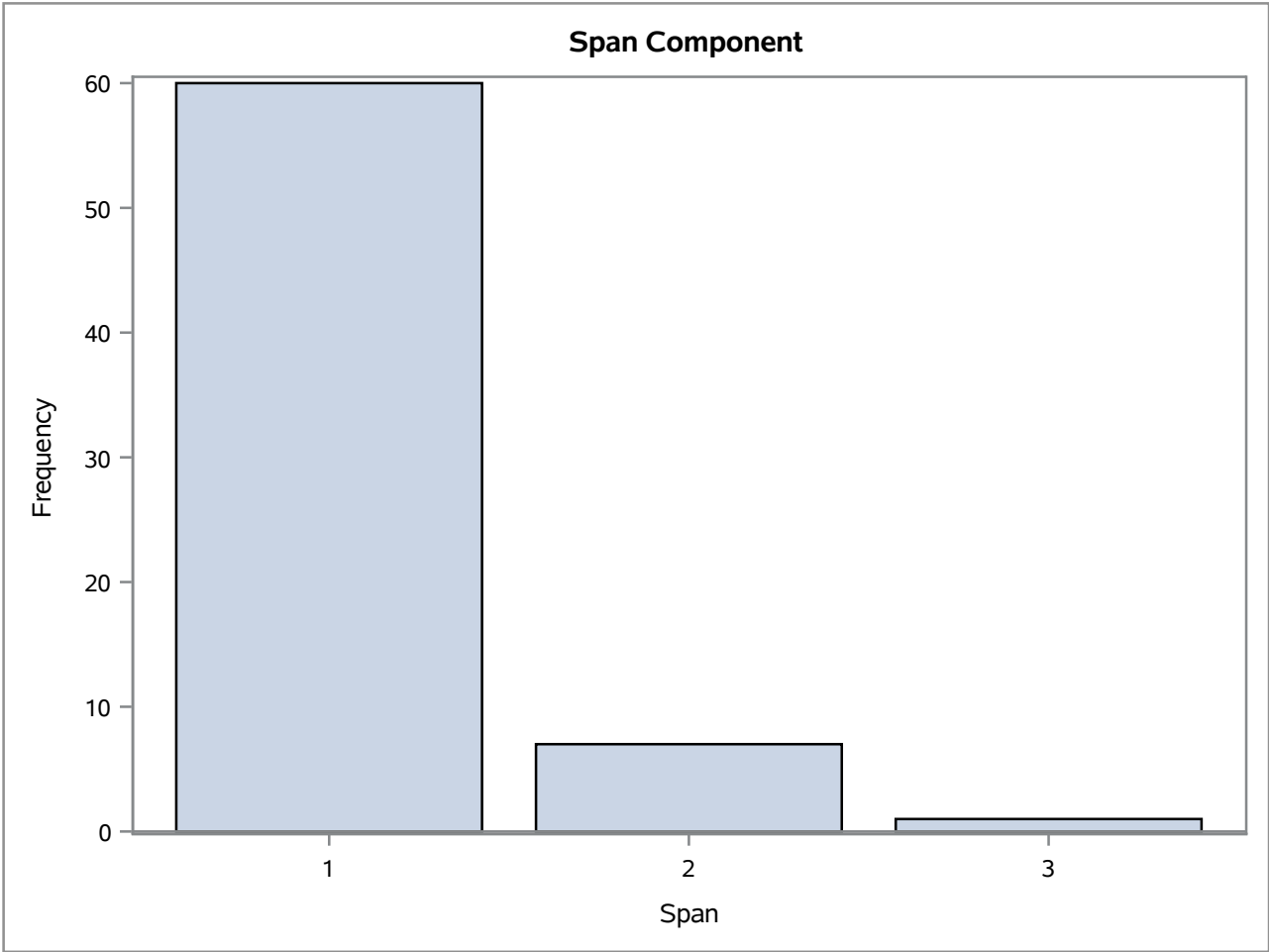
The TIMEID Procedure



Statistics Summary			
Minimum	Maximum	Mean	Standard Deviation
0	30	7.2571429	34.332644

The TIMEID Procedure

Component			
Value Index	Span	Frequency	Percentage
1	1	60	88.235294
2	2	7	10.294118
3	3	1	1.470588



Statistics Summary			
Minimum	Maximum	Mean	Standard Deviation
1	3	1.1323529	1.2425188

SECTION 2: Time Series Diagnostics
PROC TIMEID — Interval Detection, Duplicates, and Missing Periods
Dataset: Playstore | Interval: MONTH

Sunday, April 5, 2026 01:43:07 PM 9

The TIMEID Procedure

Time ID Values for Date			
Value Index	Date	Frequency	Percentage
1	JAN2012	1	1.428571
2	FEB2012	1	1.428571
3	MAR2012	1	1.428571
4	APR2012	2	2.857143
5	MAY2012	1	1.428571
6	JUN2012	1	1.428571
7	JUL2012	1	1.428571
8	AUG2012	1	1.428571
9	SEP2012	1	1.428571
10	OCT2012	1	1.428571
11	NOV2012	1	1.428571
12	DEC2012	2	2.857143
13	JAN2013	1	1.428571
14	FEB2013	1	1.428571
15	MAR2013	1	1.428571
16	MAY2013	1	1.428571
17	JUN2013	1	1.428571
18	JUL2013	1	1.428571
19	OCT2013	1	1.428571
20	NOV2013	1	1.428571
21	DEC2013	1	1.428571
22	JAN2014	1	1.428571
23	FEB2014	1	1.428571
24	MAR2014	1	1.428571
25	APR2014	1	1.428571
26	MAY2014	1	1.428571
27	JUN2014	1	1.428571
28	AUG2014	1	1.428571
29	SEP2014	1	1.428571
30	OCT2014	1	1.428571
31	NOV2014	1	1.428571
32	DEC2014	1	1.428571

Output shows time ID summary, duplicate dates, and gaps in the monthly sequence

SECTION 2: Time Series Diagnostics
PROC TIMEID — Interval Detection, Duplicates, and Missing Periods
Dataset: Playstore | Interval: MONTH

Sunday, April 5, 2026 01:43:07 PM 10

The TIMEID Procedure

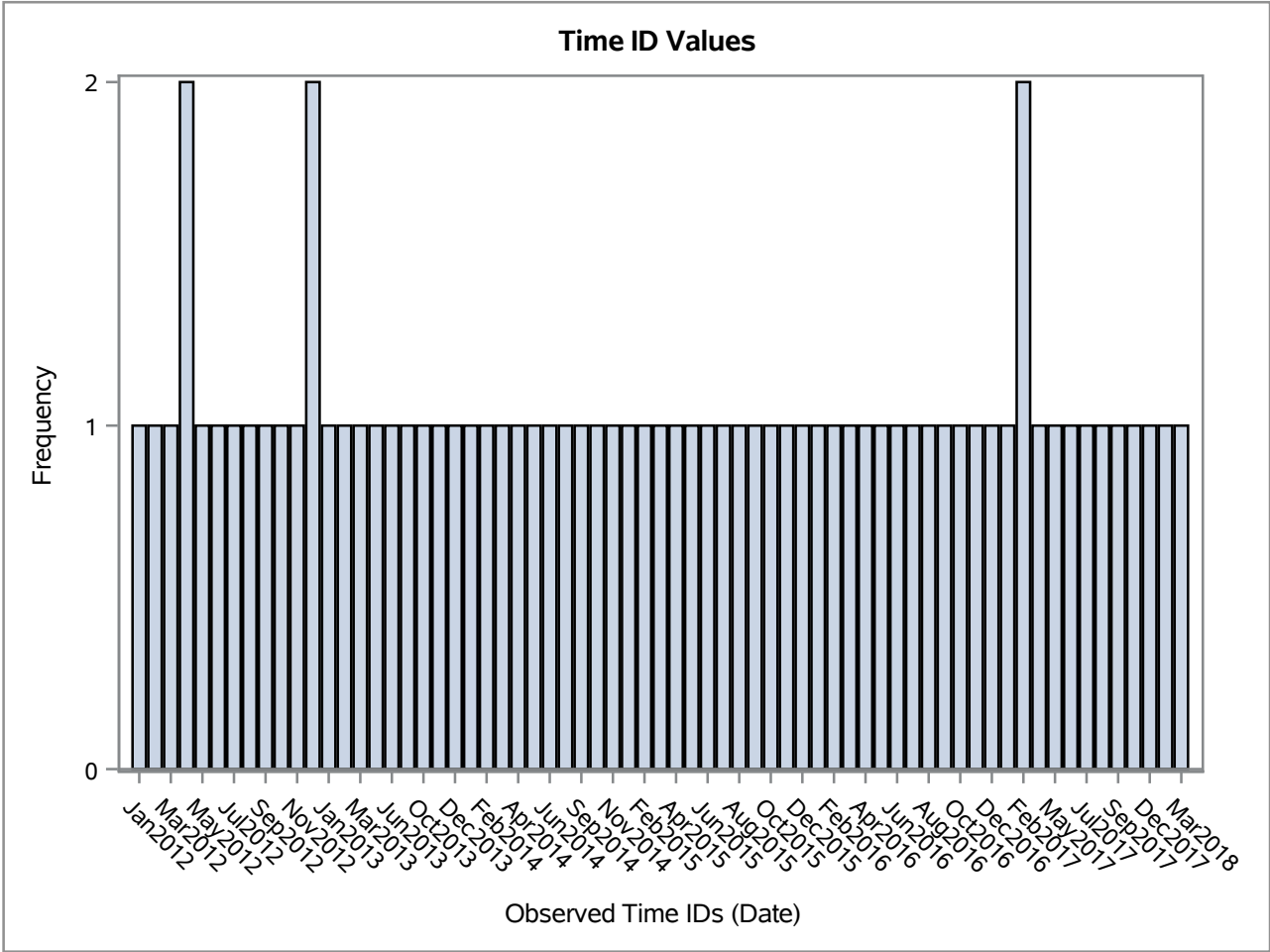
Time ID Values for Date			
Value Index	Date	Frequency	Percentage
33	FEB2015	1	1.428571
34	MAR2015	1	1.428571
35	APR2015	1	1.428571
36	MAY2015	1	1.428571
37	JUN2015	1	1.428571
38	JUL2015	1	1.428571
39	AUG2015	1	1.428571
40	SEP2015	1	1.428571
41	OCT2015	1	1.428571
42	NOV2015	1	1.428571
43	DEC2015	1	1.428571
44	JAN2016	1	1.428571
45	FEB2016	1	1.428571
46	MAR2016	1	1.428571
47	APR2016	1	1.428571
48	MAY2016	1	1.428571
49	JUN2016	1	1.428571
50	JUL2016	1	1.428571
51	AUG2016	1	1.428571
52	SEP2016	1	1.428571
53	OCT2016	1	1.428571
54	NOV2016	1	1.428571
55	DEC2016	1	1.428571
56	JAN2017	1	1.428571
57	FEB2017	2	2.857143
58	APR2017	1	1.428571
59	MAY2017	1	1.428571
60	JUN2017	1	1.428571
61	JUL2017	1	1.428571
62	AUG2017	1	1.428571
63	SEP2017	1	1.428571
64	NOV2017	1	1.428571

Output shows time ID summary, duplicate dates, and gaps in the monthly sequence

The TIMEID Procedure

Time ID Values for Date			
Value Index	Date	Frequency	Percentage
65	DEC2017	1	1.428571
66	FEB2018	1	1.428571
67	MAR2018	1	1.428571

Time ID Value Summary for Date				
N Obs Excluded	N Obs Read	N Obs Missing	N Obs Used	Unique Values
0	70	0	70	67



Output shows time ID summary, duplicate dates, and gaps in the monthly sequence

The TIMEID Procedure

Time Interval Analysis Summary	
Time ID Variable	Date
Time Interval	MONTH
Base Name	MONTH
Multiplier	1
Shift	0
Length of Seasonal Cycle	12
Time ID Format	MONYY7.
Start	JAN2012
End	MAR2018

Output shows time ID summary, duplicate dates, and gaps in the monthly sequence

SECTION 4: Dataset Inspection
PROC CONTENTS — Variable Names, Types, and Structure
Dataset: WORK.inflation

Sunday, April 5, 2026 01:43:07 PM 13

The CONTENTS Procedure

Data Set Name	WORK.INFLATION	Observations	75
Member Type	DATA	Variables	11
Engine	V9	Indexes	0
Created	04/05/2026 13:43:08	Observation Length	88
Last Modified	04/05/2026 13:43:08	Deleted Observations	0
Protection		Compressed	NO
Data Set Type		Sorted	NO
Label			
Data Representation	SOLARIS_X86_64, LINUX_X86_64, ALPHA_TRU64, LINUX_IA64		
Encoding	utf-8 Unicode (UTF-8)		

Engine/Host Dependent Information	
Data Set Page Size	131072
Number of Data Set Pages	1
First Data Page	1
Max Obs per Page	1486
Obs in First Data Page	75
Number of Data Set Repairs	0
Filename	/saswork/SAS_work071D0000EAD1_odaws01-usw2-2.oda.sas.com/SAS_workD6910000EAD1_odaws01-usw2-2.oda.sas.com/inflation.sas7bdat
Release Created	9.0401M8
Host Created	Linux
Inode Number	2483029644
Access Permission	rw-r--r--
Owner Name	u64366633
File Size	256KB
File Size (bytes)	262144

Alphabetic List of Variables and Attributes					
#	Variable	Type	Len	Format	Label
11	CPI	Num	8	BEST.	CPI
9	Clothing_and_shopping	Num	8	BEST.	Clothing_and_shopping
8	Communication	Num	8	BEST.	Communication
4	Eating_out	Num	8	BEST.	Eating_out

Confirms all variables were imported correctly from the Excel source file

SECTION 4: Dataset Inspection
PROC CONTENTS — Variable Names, Types, and Structure
Dataset: WORK.inflation

Sunday, April 5, 2026 01:43:07 PM 14

The CONTENTS Procedure

Alphabetic List of Variables and Attributes					
#	Variable	Type	Len	Format	Label
7	Education	Num	8	BEST.	Education
5	Entertainment	Num	8	BEST.	Entertainment
2	Furniture_Home_Improvement	Num	8	BEST.	Furniture_Home_Improvement
6	Grocery	Num	8	BEST.	Grocery
1	Month	Num	8	MONYY7.	Month
10	Spend_save_quaterly_ratio	Num	8	BEST.	Spend_save_quaterly_ratio
3	Travel_including_Leisure	Num	8	BEST.	Travel_including_Leisure

Confirms all variables were imported correctly from the Excel source file

**SECTION 5 (VERIFICATION): Last 10 Rows of Model Dataset
Confirms CPI Is Missing for the Hold-Out Period**

Obs	Month	CPI
60	DEC2016	106.2
61	JAN2017	106.2
62	FEB2017	106.2
63	MAR2017	106.2
64	APR2017	106.2
65	MAY2017	106.3
66	JUN2017	106.3
67	JUL2017	106.2
68	AUG2017	106.3
69	SEP2017	106.4
70	OCT2017	.

Rows after 30SEP2017 should show CPI = .

SECTION 6A: Multivariate Regression — Forward Selection
Starts with no variables, adds the best predictor at each step
Response: CPI | Predictors: 9 Consumer Spending Categories

The REG Procedure
Model: MODEL1
Dependent Variable: CPI CPI

Number of Observations Read	75
Number of Observations Used	69
Number of Observations with Missing Values	6

Forward Selection: Step 1

Statistics for Entry DF = 1,67				
Variable	Tolerance	Model R-Square	F Value	Pr > F
Furniture_Home_Improvement	1.000000	0.2405	21.21	<.0001
Travel_including_Leisure	1.000000	0.5712	89.27	<.0001
Eating_out	1.000000	0.6670	134.20	<.0001
Entertainment	1.000000	0.4943	65.50	<.0001
Grocery	1.000000	0.6629	131.73	<.0001
Education	1.000000	0.8195	304.24	<.0001
Communication	1.000000	0.0281	1.93	0.1689
Clothing_and_shopping	1.000000	0.8239	313.54	<.0001
Spend_save_quaterly_ratio	1.000000	0.8661	433.27	<.0001

Variable Spend_save_quaterly_ratio Entered: R-Square = 0.8661 and C(p) = 101.1809

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	1	18.25606	18.25606	433.27	<.0001
Error	67	2.82307	0.04214		
Corrected Total	68	21.07913			

Variable	Parameter Estimate	Standard Error	Type II SS	F Value	Pr > F
Intercept	91.90309	0.64805	847.39683	20111.3	<.0001
Spend_save_quaterly_ratio	0.12923	0.00621	18.25606	433.27	<.0001

Bounds on condition number: 1, 1

Selection criteria: SBC | Details=All shows step-by-step entry log

SECTION 6A: Multivariate Regression — Forward Selection
Starts with no variables, adds the best predictor at each step
Response: CPI | Predictors: 9 Consumer Spending Categories

The REG Procedure
Model: MODEL1
Dependent Variable: CPI CPI

Forward Selection: Step 2

Statistics for Entry DF = 1,66				
Variable	Tolerance	Model R-Square	F Value	Pr > F
Furniture_Home_Improvement	0.689061	0.8673	0.59	0.4457
Travel_including_Leisure	0.347601	0.8661	0.02	0.8770
Eating_out	0.265277	0.8674	0.68	0.4133
Entertainment	0.355325	0.8716	2.82	0.0980
Grocery	0.228051	0.8661	0.03	0.8715
Education	0.148517	0.8806	8.06	0.0060
Communication	0.944481	0.8689	1.43	0.2362
Clothing_and_shopping	0.240857	0.9050	27.07	<.0001

Variable Clothing_and_shopping Entered: R-Square = 0.9050 and C(p) = 54.8481

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	2	19.07714	9.53857	314.46	<.0001
Error	66	2.00199	0.03033		
Corrected Total	68	21.07913			

Variable	Parameter Estimate	Standard Error	Type II SS	F Value	Pr > F
Intercept	85.53920	1.34109	123.40594	4068.34	<.0001
Clothing_and_shopping	0.11089	0.02131	0.82107	27.07	<.0001
Spend_save_quaterly_ratio	0.08058	0.01073	1.70936	56.35	<.0001

Bounds on condition number: 4.1518, 16.607

SECTION 6A: Multivariate Regression — Forward Selection
Starts with no variables, adds the best predictor at each step
Response: CPI | Predictors: 9 Consumer Spending Categories

The REG Procedure
Model: MODEL1
Dependent Variable: CPI CPI

Forward Selection: Step 3

Statistics for Entry DF = 1,65				
Variable	Tolerance	Model R-Square	F Value	Pr > F
Furniture_Home_Improvement	0.481411	0.9342	28.81	<.0001
Travel_including_Leisure	0.198806	0.9311	24.60	<.0001
Eating_out	0.247564	0.9052	0.15	0.7035
Entertainment	0.285849	0.9374	33.57	<.0001
Grocery	0.186638	0.9152	7.83	0.0068
Education	0.068038	0.9063	0.91	0.3428
Communication	0.940321	0.9066	1.13	0.2920

Variable Entertainment Entered: R-Square = 0.9374 and C(p) = 16.7114

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	3	19.75898	6.58633	324.29	<.0001
Error	65	1.32016	0.02031		
Corrected Total	68	21.07913			

Variable	Parameter Estimate	Standard Error	Type II SS	F Value	Pr > F
Intercept	83.43388	1.15596	105.80571	5209.52	<.0001
Entertainment	-0.04118	0.00711	0.68184	33.57	<.0001
Clothing_and_shopping	0.16070	0.01944	1.38735	68.31	<.0001
Spend_save_quaterly_ratio	0.09622	0.00919	2.22716	109.66	<.0001

Bounds on condition number: 5.161, 39.611

SECTION 6A: Multivariate Regression — Forward Selection
Starts with no variables, adds the best predictor at each step
Response: CPI | Predictors: 9 Consumer Spending Categories

The REG Procedure
Model: MODEL1
Dependent Variable: CPI CPI

Forward Selection: Step 4

Statistics for Entry DF = 1,64				
Variable	Tolerance	Model R-Square	F Value	Pr > F
Furniture_Home_Improvement	0.127589	0.9384	1.08	0.3026
Travel_including_Leisure	0.050971	0.9375	0.16	0.6873
Eating_out	0.239789	0.9377	0.32	0.5757
Grocery	0.101851	0.9381	0.77	0.3833
Education	0.036368	0.9513	18.38	<.0001
Communication	0.938853	0.9385	1.15	0.2885

Variable Education Entered: R-Square = 0.9513 and C(p) = 1.3699

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	4	20.05357	5.01339	312.86	<.0001
Error	64	1.02556	0.01602		
Corrected Total	68	21.07913			

Variable	Parameter Estimate	Standard Error	Type II SS	F Value	Pr > F
Intercept	84.33993	1.04830	103.72301	6472.84	<.0001
Entertainment	-0.06644	0.00863	0.94870	59.20	<.0001
Education	0.11236	0.02620	0.29460	18.38	<.0001
Clothing_and_shopping	0.09246	0.02349	0.24830	15.50	0.0002
Spend_save_quaterly_ratio	0.06974	0.01024	0.74399	46.43	<.0001

Bounds on condition number: 27.497, 202.93

SECTION 6A: Multivariate Regression — Forward Selection
Starts with no variables, adds the best predictor at each step
Response: CPI | Predictors: 9 Consumer Spending Categories

The REG Procedure
Model: MODEL1
Dependent Variable: CPI CPI

Forward Selection: Step 5

Statistics for Entry DF = 1,63				
Variable	Tolerance	Model R-Square	F Value	Pr > F
Furniture_Home_Improvement	0.105777	0.9517	0.44	0.5117
Travel_including_Leisure	0.050890	0.9516	0.39	0.5330
Eating_out	0.239778	0.9516	0.37	0.5477
Grocery	0.096775	0.9513	0.00	0.9734
Communication	0.918195	0.9516	0.33	0.5701

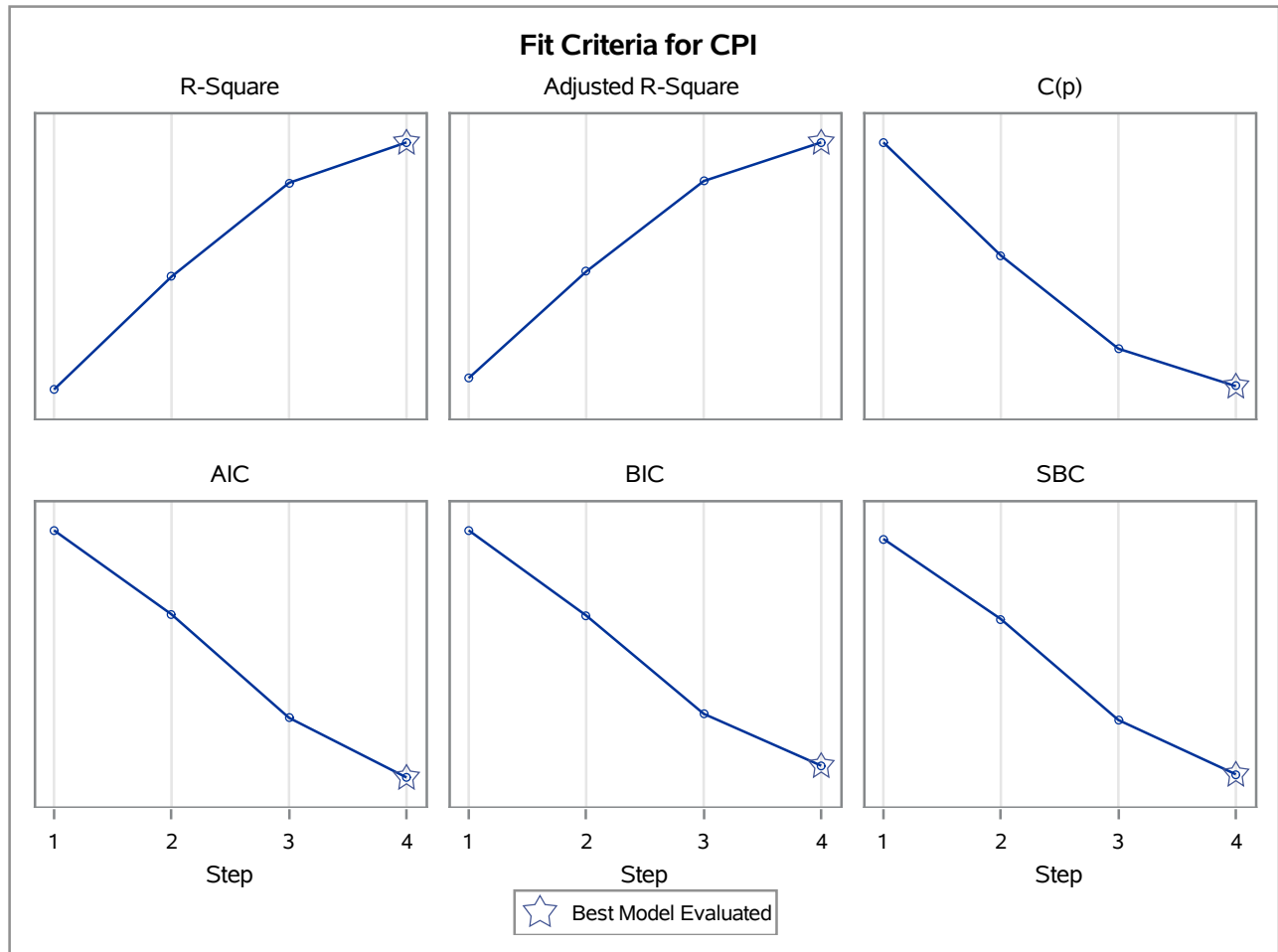
No other variable met the 0.5000 significance level for entry into the model.

Summary of Forward Selection								
Step	Variable Entered	Label	Number Vars In	Partial R-Square	Model R-Square	C(p)	F Value	Pr > F
1	Spend_save_quaterly_ratio	Spend_save_quaterly_ratio	1	0.8661	0.8661	101.181	433.27	<.0001
2	Clothing_and_shopping	Clothing_and_shopping	2	0.0390	0.9050	54.8481	27.07	<.0001
3	Entertainment	Entertainment	3	0.0323	0.9374	16.7114	33.57	<.0001
4	Education	Education	4	0.0140	0.9513	1.3699	18.38	<.0001

Selection criteria: SBC | Details=All shows step-by-step entry log

SECTION 6A: Multivariate Regression — Forward Selection
Starts with no variables, adds the best predictor at each step
Response: CPI | Predictors: 9 Consumer Spending Categories

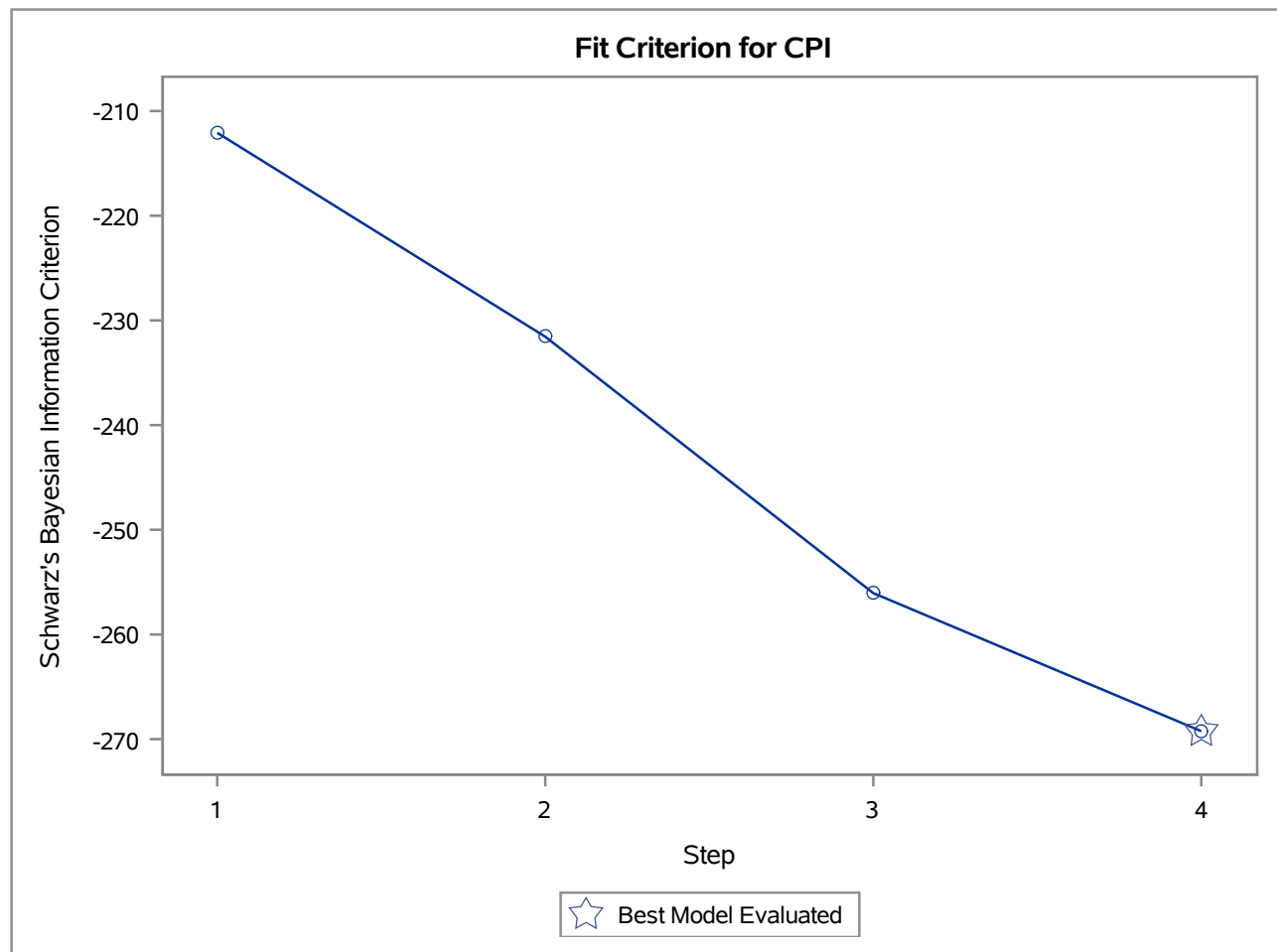
The REG Procedure
Model: MODEL1
Dependent Variable: CPI CPI



Selection criteria: SBC | Details=All shows step-by-step entry log

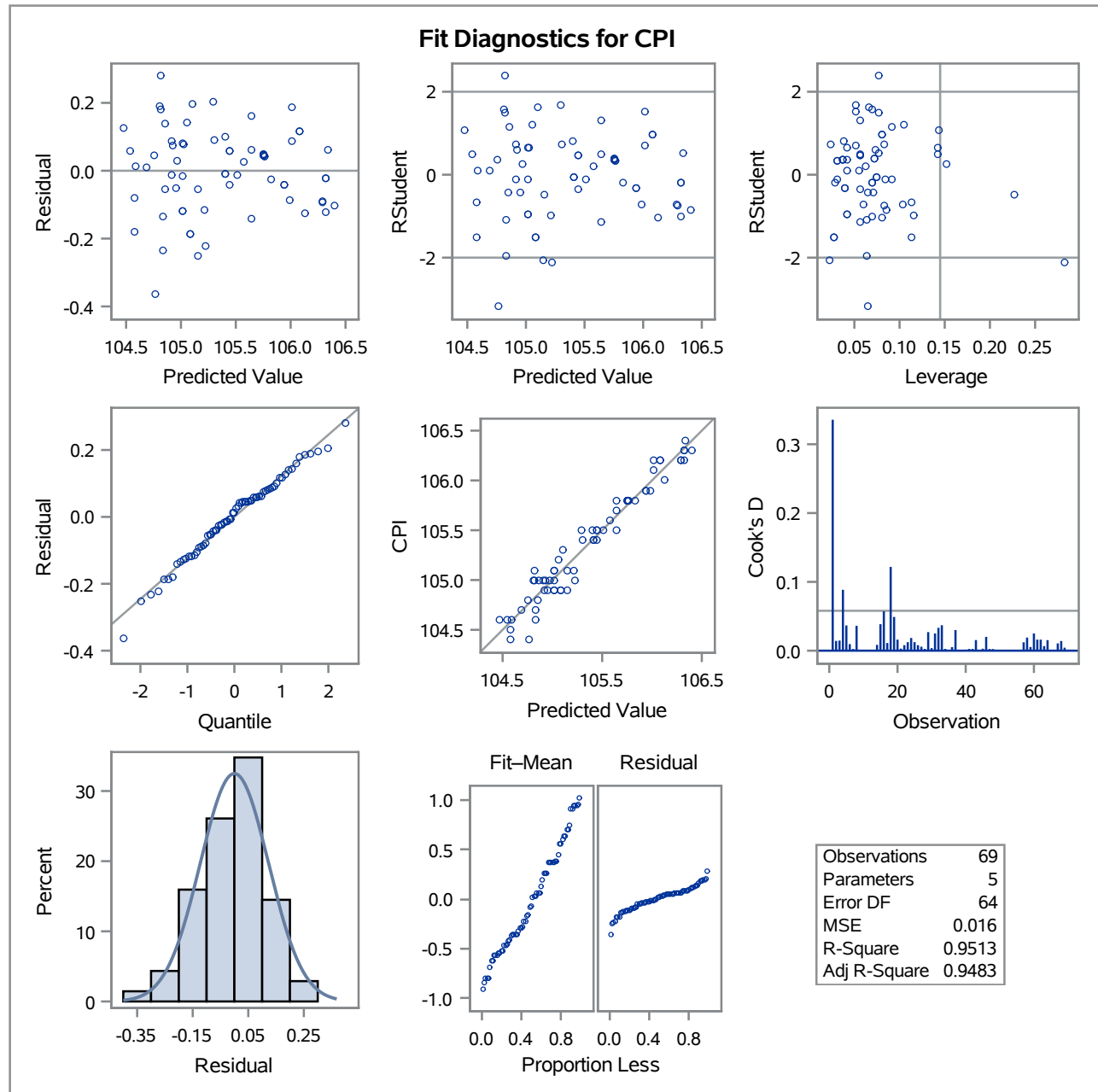
SECTION 6A: Multivariate Regression — Forward Selection
Starts with no variables, adds the best predictor at each step
Response: CPI | Predictors: 9 Consumer Spending Categories

The REG Procedure
Model: MODEL1
Dependent Variable: CPI CPI



SECTION 6A: Multivariate Regression — Forward Selection
Starts with no variables, adds the best predictor at each step
Response: CPI | Predictors: 9 Consumer Spending Categories

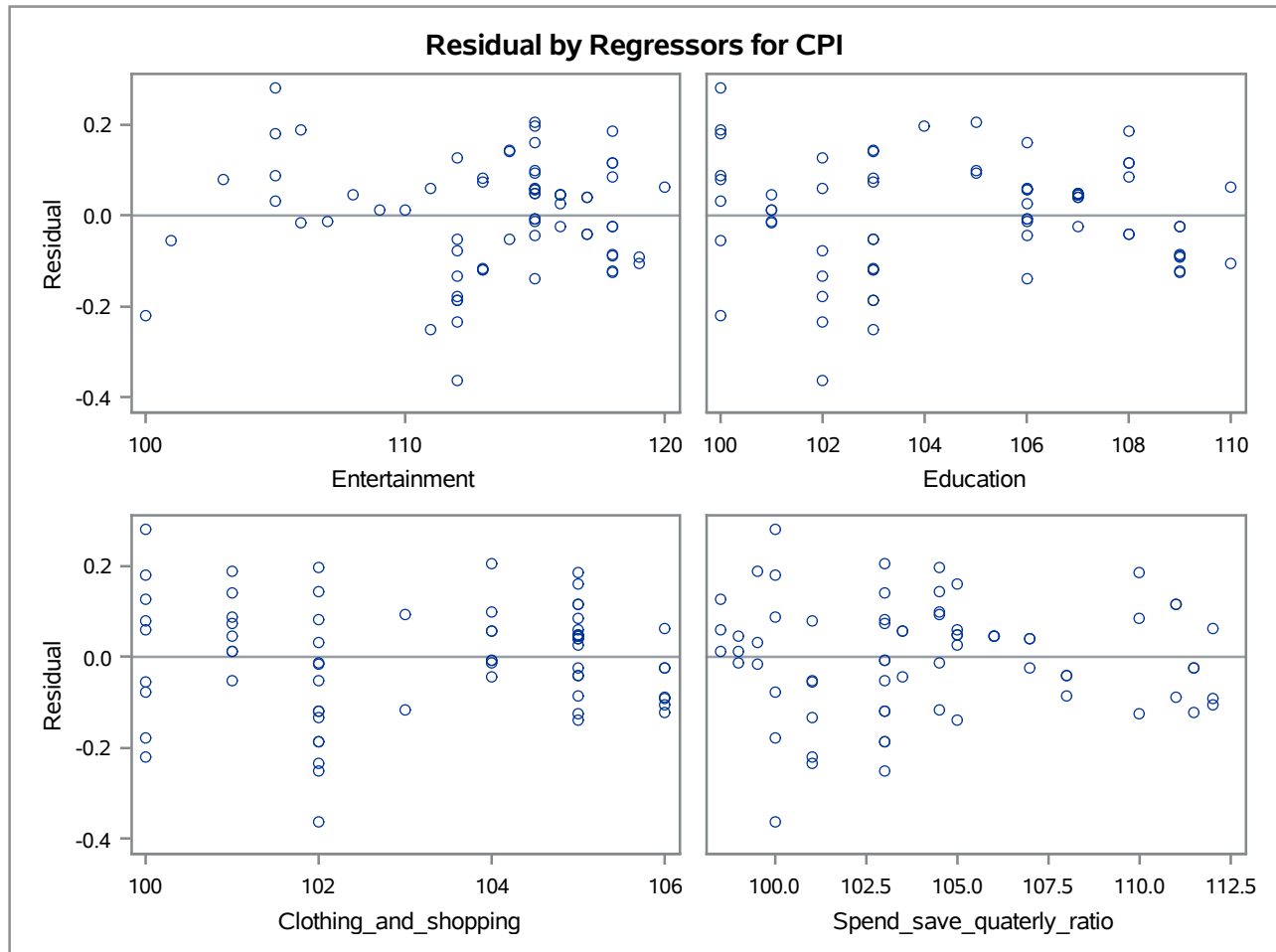
The REG Procedure
Model: MODEL1
Dependent Variable: CPI CPI



Selection criteria: SBC | Details=All shows step-by-step entry log

SECTION 6A: Multivariate Regression — Forward Selection
Starts with no variables, adds the best predictor at each step
Response: CPI | Predictors: 9 Consumer Spending Categories

The REG Procedure
Model: MODEL1
Dependent Variable: CPI CPI



SECTION 6B: Multivariate Regression — Backward Selection
Starts with all 9 variables, removes the weakest predictor at each step
Response: CPI | Predictors: 9 Consumer Spending Categories

The REG Procedure
Model: MODEL1
Dependent Variable: CPI CPI

Number of Observations Read	75
Number of Observations Used	69
Number of Observations with Missing Values	6

Backward Elimination: Step 0

All Variables Entered: R-Square = 0.9525 and C(p) = 10.0000

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	9	20.07684	2.23076	131.31	<.0001
Error	59	1.00229	0.01699		
Corrected Total	68	21.07913			

Variable	Parameter Estimate	Standard Error	Type II SS	F Value	Pr > F
Intercept	83.13330	3.04430	12.66819	745.72	<.0001
Furniture_Home_Improvement	0.00911	0.01382	0.00738	0.43	0.5125
Travel_including_Leisure	-0.00750	0.01098	0.00793	0.47	0.4972
Eating_out	-0.00015179	0.01061	0.00000348	0.00	0.9886
Entertainment	-0.06756	0.01989	0.19602	11.54	0.0012
Grocery	0.00123	0.03527	0.00002060	0.00	0.9723
Education	0.11989	0.03200	0.23849	14.04	0.0004
Communication	-0.00094288	0.00152	0.00656	0.39	0.5367
Clothing_and_shopping	0.09299	0.03123	0.15059	8.86	0.0042
Spend_save_quaterly_ratio	0.07273	0.01410	0.45186	26.60	<.0001

Bounds on condition number: 38.673, 1478

Selection criteria: SBC | Details=All shows step-by-step removal log

SECTION 6B: Multivariate Regression — Backward Selection
Starts with all 9 variables, removes the weakest predictor at each step
Response: CPI | Predictors: 9 Consumer Spending Categories

The REG Procedure
Model: MODEL1
Dependent Variable: CPI CPI

Backward Elimination: Step 1

Statistics for Removal DF = 1,59				
Variable	Partial R-Square	Model R-Square	F Value	Pr > F
Furniture_Home_Improvement	0.0004	0.9521	0.43	0.5125
Travel_including_Leisure	0.0004	0.9521	0.47	0.4972
Eating_out	0.0000	0.9525	0.00	0.9886
Entertainment	0.0093	0.9432	11.54	0.0012
Grocery	0.0000	0.9525	0.00	0.9723
Education	0.0113	0.9411	14.04	0.0004
Communication	0.0003	0.9521	0.39	0.5367
Clothing_and_shopping	0.0071	0.9453	8.86	0.0042
Spend_save_quaterly_ratio	0.0214	0.9310	26.60	<.0001

Variable Eating_out Removed: R-Square = 0.9525 and C(p) = 8.0002

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	8	20.07684	2.50960	150.23	<.0001
Error	60	1.00229	0.01670		
Corrected Total	68	21.07913			

Variable	Parameter Estimate	Standard Error	Type II SS	F Value	Pr > F
Intercept	83.14480	2.91172	13.62114	815.40	<.0001
Furniture_Home_Improvement	0.00901	0.01205	0.00934	0.56	0.4576
Travel_including_Leisure	-0.00741	0.00878	0.01188	0.71	0.4024
Entertainment	-0.06759	0.01959	0.19883	11.90	0.0010
Grocery	0.00130	0.03461	0.00002364	0.00	0.9701
Education	0.11976	0.03049	0.25767	15.42	0.0002
Communication	-0.00094349	0.00150	0.00657	0.39	0.5328
Clothing_and_shopping	0.09292	0.03056	0.15446	9.25	0.0035
Spend_save_quaterly_ratio	0.07263	0.01205	0.60671	36.32	<.0001

Selection criteria: SBC | Details=All shows step-by-step removal log

SECTION 6B: Multivariate Regression — Backward Selection
Starts with all 9 variables, removes the weakest predictor at each step
Response: CPI | Predictors: 9 Consumer Spending Categories

The REG Procedure
Model: MODEL1
Dependent Variable: CPI CPI

Backward Elimination: Step 1

Bounds on condition number: 35.718, 1083.5

Backward Elimination: Step 2

Statistics for Removal DF = 1,60				
Variable	Partial R-Square	Model R-Square	F Value	Pr > F
Furniture_Home_Improvement	0.0004	0.9520	0.56	0.4576
Travel_including_Leisure	0.0006	0.9519	0.71	0.4024
Entertainment	0.0094	0.9430	11.90	0.0010
Grocery	0.0000	0.9524	0.00	0.9701
Education	0.0122	0.9402	15.42	0.0002
Communication	0.0003	0.9521	0.39	0.5328
Clothing_and_shopping	0.0073	0.9451	9.25	0.0035
Spend_save_quaterly_ratio	0.0288	0.9237	36.32	<.0001

Variable Grocery Removed: R-Square = 0.9524 and C(p) = 6.0016

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	7	20.07682	2.86812	174.55	<.0001
Error	61	1.00231	0.01643		
Corrected Total	68	21.07913			

Variable	Parameter Estimate	Standard Error	Type II SS	F Value	Pr > F
Intercept	83.23078	1.78902	35.56396	2164.39	<.0001
Furniture_Home_Improvement	0.00901	0.01195	0.00933	0.57	0.4539
Travel_including_Leisure	-0.00739	0.00869	0.01186	0.72	0.3988
Entertainment	-0.06744	0.01901	0.20669	12.58	0.0008
Education	0.11998	0.02969	0.26828	16.33	0.0002
Communication	-0.00094741	0.00149	0.00666	0.41	0.5267
Clothing_and_shopping	0.09286	0.03027	0.15465	9.41	0.0032
Spend_save_quaterly_ratio	0.07274	0.01155	0.65162	39.66	<.0001

Selection criteria: SBC | Details=All shows step-by-step removal log

SECTION 6B: Multivariate Regression — Backward Selection
Starts with all 9 variables, removes the weakest predictor at each step
Response: CPI | Predictors: 9 Consumer Spending Categories

The REG Procedure
Model: MODEL1
Dependent Variable: CPI CPI

Backward Elimination: Step 2

Bounds on condition number: 34.431, 850.97

Backward Elimination: Step 3

Statistics for Removal DF = 1,61				
Variable	Partial R-Square	Model R-Square	F Value	Pr > F
Furniture_Home_Improvement	0.0004	0.9520	0.57	0.4539
Travel_including_Leisure	0.0006	0.9519	0.72	0.3988
Entertainment	0.0098	0.9426	12.58	0.0008
Education	0.0127	0.9397	16.33	0.0002
Communication	0.0003	0.9521	0.41	0.5267
Clothing_and_shopping	0.0073	0.9451	9.41	0.0032
Spend_save_quaterly_ratio	0.0309	0.9215	39.66	<.0001

Variable Communication Removed: R-Square = 0.9521 and C(p) = 4.3937

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	6	20.07016	3.34503	205.55	<.0001
Error	62	1.00898	0.01627		
Corrected Total	68	21.07913			

Variable	Parameter Estimate	Standard Error	Type II SS	F Value	Pr > F
Intercept	83.31267	1.77582	35.81913	2201.03	<.0001
Furniture_Home_Improvement	0.00942	0.01188	0.01023	0.63	0.4310
Travel_including_Leisure	-0.00655	0.00855	0.00954	0.59	0.4467
Entertainment	-0.06974	0.01858	0.22934	14.09	0.0004
Education	0.12287	0.02920	0.28805	17.70	<.0001
Clothing_and_shopping	0.09037	0.02987	0.14894	9.15	0.0036
Spend_save_quaterly_ratio	0.07167	0.01137	0.64629	39.71	<.0001

Selection criteria: SBC | Details=All shows step-by-step removal log

SECTION 6B: Multivariate Regression — Backward Selection
Starts with all 9 variables, removes the weakest predictor at each step
Response: CPI | Predictors: 9 Consumer Spending Categories

The REG Procedure
Model: MODEL1
Dependent Variable: CPI CPI

Backward Elimination: Step 3

Bounds on condition number: 33.629, 705.41

Backward Elimination: Step 4

Statistics for Removal DF = 1,62				
Variable	Partial R-Square	Model R-Square	F Value	Pr > F
Furniture_Home_Improvement	0.0005	0.9516	0.63	0.4310
Travel_including_Leisure	0.0005	0.9517	0.59	0.4467
Entertainment	0.0109	0.9413	14.09	0.0004
Education	0.0137	0.9385	17.70	<.0001
Clothing_and_shopping	0.0071	0.9451	9.15	0.0036
Spend_save_quaterly_ratio	0.0307	0.9215	39.71	<.0001

Variable Travel_including_Leisure Removed: R-Square = 0.9517 and C(p) = 2.9554

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	5	20.06061	4.01212	248.17	<.0001
Error	63	1.01852	0.01617		
Corrected Total	68	21.07913			

Variable	Parameter Estimate	Standard Error	Type II SS	F Value	Pr > F
Intercept	84.40323	1.05731	103.02395	6372.50	<.0001
Furniture_Home_Improvement	0.00767	0.01162	0.00704	0.44	0.5117
Entertainment	-0.07587	0.01671	0.33320	20.61	<.0001
Education	0.12025	0.02891	0.27973	17.30	<.0001
Clothing_and_shopping	0.08254	0.02797	0.14074	8.71	0.0044
Spend_save_quaterly_ratio	0.07275	0.01125	0.67641	41.84	<.0001

Bounds on condition number: 33.167, 444.45

Selection criteria: SBC | Details=All shows step-by-step removal log

SECTION 6B: Multivariate Regression — Backward Selection
Starts with all 9 variables, removes the weakest predictor at each step
Response: CPI | Predictors: 9 Consumer Spending Categories

The REG Procedure
Model: MODEL1
Dependent Variable: CPI CPI

Backward Elimination: Step 5

Statistics for Removal DF = 1,63				
Variable	Partial R-Square	Model R-Square	F Value	Pr > F
Furniture_Home_Improvement	0.0003	0.9513	0.44	0.5117
Entertainment	0.0158	0.9359	20.61	<.0001
Education	0.0133	0.9384	17.30	<.0001
Clothing_and_shopping	0.0067	0.9450	8.71	0.0044
Spend_save_quaterly_ratio	0.0321	0.9196	41.84	<.0001

Variable Furniture_Home_Improvement Removed: R-Square = 0.9513 and C(p) = 1.3699

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	4	20.05357	5.01339	312.86	<.0001
Error	64	1.02556	0.01602		
Corrected Total	68	21.07913			

Variable	Parameter Estimate	Standard Error	Type II SS	F Value	Pr > F
Intercept	84.33993	1.04830	103.72301	6472.84	<.0001
Entertainment	-0.06644	0.00863	0.94870	59.20	<.0001
Education	0.11236	0.02620	0.29460	18.38	<.0001
Clothing_and_shopping	0.09246	0.02349	0.24830	15.50	0.0002
Spend_save_quaterly_ratio	0.06974	0.01024	0.74399	46.43	<.0001

Bounds on condition number: 27.497, 202.93

SECTION 6B: Multivariate Regression — Backward Selection
Starts with all 9 variables, removes the weakest predictor at each step
Response: CPI | Predictors: 9 Consumer Spending Categories

The REG Procedure
Model: MODEL1
Dependent Variable: CPI CPI

Backward Elimination: Step 6

Statistics for Removal DF = 1,64				
Variable	Partial R-Square	Model R-Square	F Value	Pr > F
Entertainment	0.0450	0.9063	59.20	<.0001
Education	0.0140	0.9374	18.38	<.0001
Clothing_and_shopping	0.0118	0.9396	15.50	0.0002
Spend_save_quaterly_ratio	0.0353	0.9161	46.43	<.0001

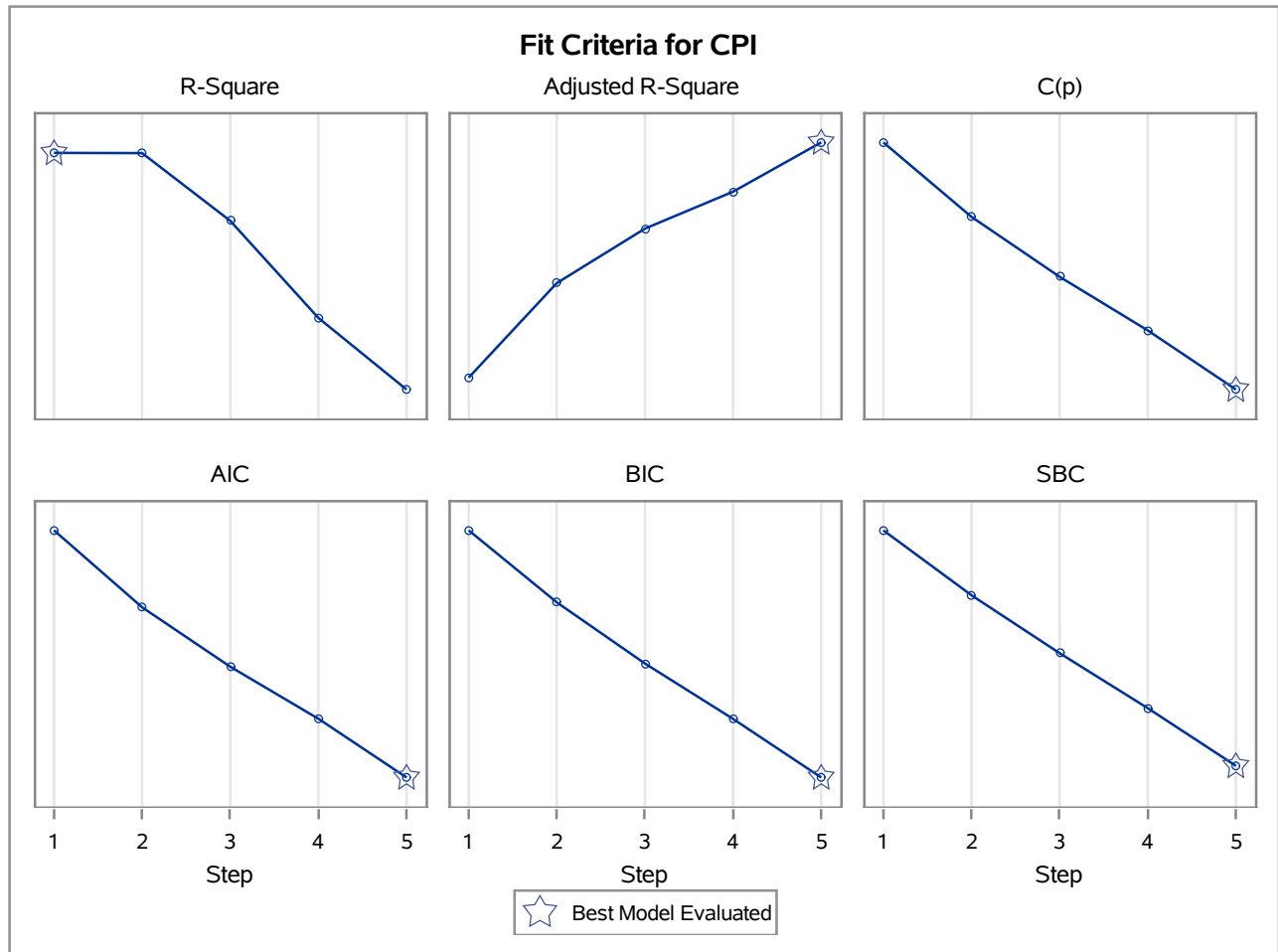
All variables left in the model are significant at the 0.1000 level.

Summary of Backward Elimination								
Step	Variable Removed	Label	Number Vars In	Partial R-Square	Model R-Square	C(p)	F Value	Pr > F
1	Eating_out	Eating_out	8	0.0000	0.9525	8.0002	0.00	0.9886
2	Grocery	Grocery	7	0.0000	0.9524	6.0016	0.00	0.9701
3	Communication	Communication	6	0.0003	0.9521	4.3937	0.41	0.5267
4	Travel_including_Leisure	Travel_including_Leisure	5	0.0005	0.9517	2.9554	0.59	0.4467
5	Furniture_Home_Improvement	Furniture_Home_Improvement	4	0.0003	0.9513	1.3699	0.44	0.5117

Selection criteria: SBC | Details=All shows step-by-step removal log

SECTION 6B: Multivariate Regression — Backward Selection
Starts with all 9 variables, removes the weakest predictor at each step
Response: CPI | Predictors: 9 Consumer Spending Categories

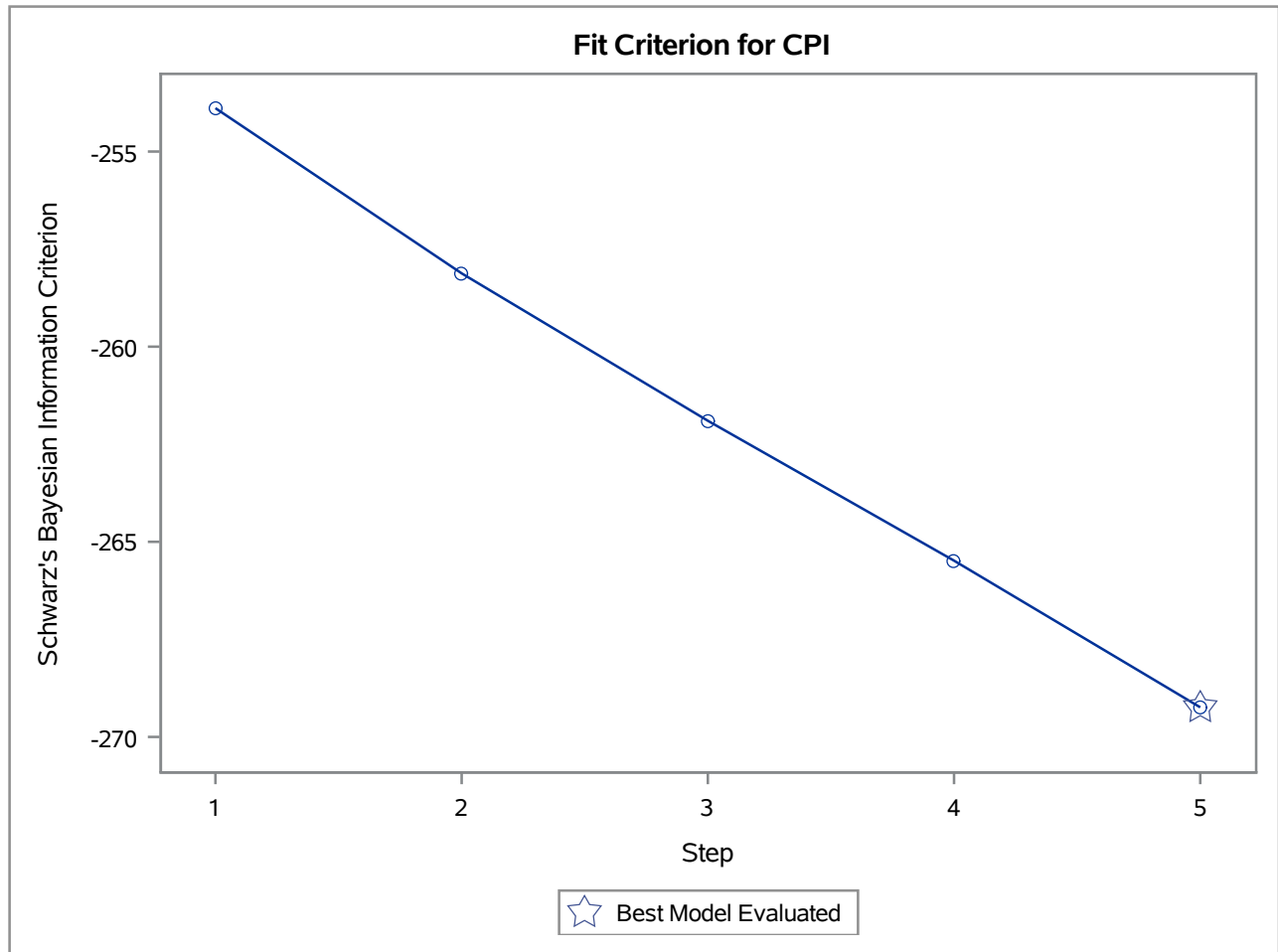
The REG Procedure
Model: MODEL1
Dependent Variable: CPI CPI



Selection criteria: SBC | Details=All shows step-by-step removal log

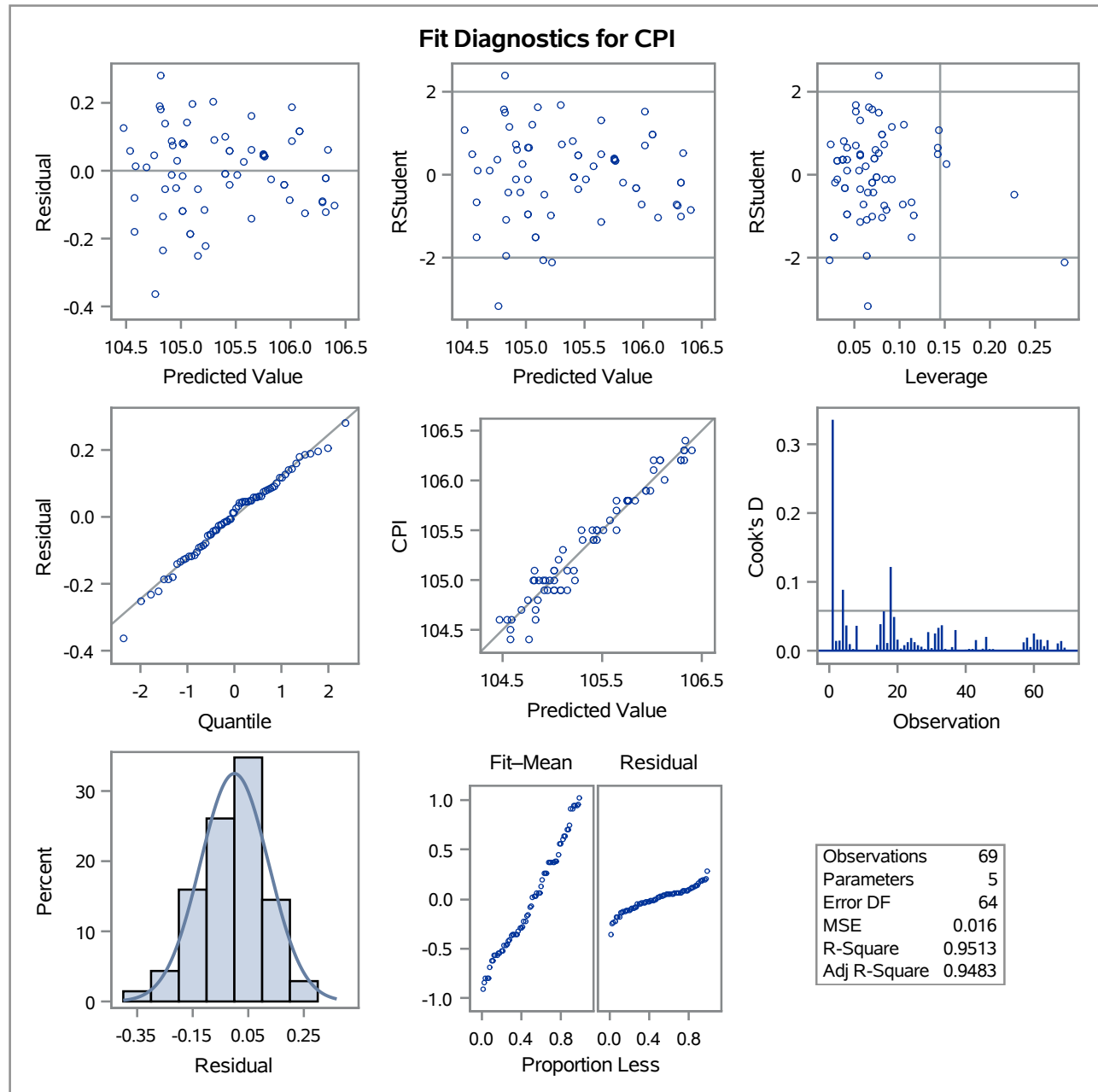
SECTION 6B: Multivariate Regression — Backward Selection
Starts with all 9 variables, removes the weakest predictor at each step
Response: CPI | Predictors: 9 Consumer Spending Categories

The REG Procedure
 Model: MODEL1
 Dependent Variable: CPI CPI



SECTION 6B: Multivariate Regression — Backward Selection
Starts with all 9 variables, removes the weakest predictor at each step
Response: CPI | Predictors: 9 Consumer Spending Categories

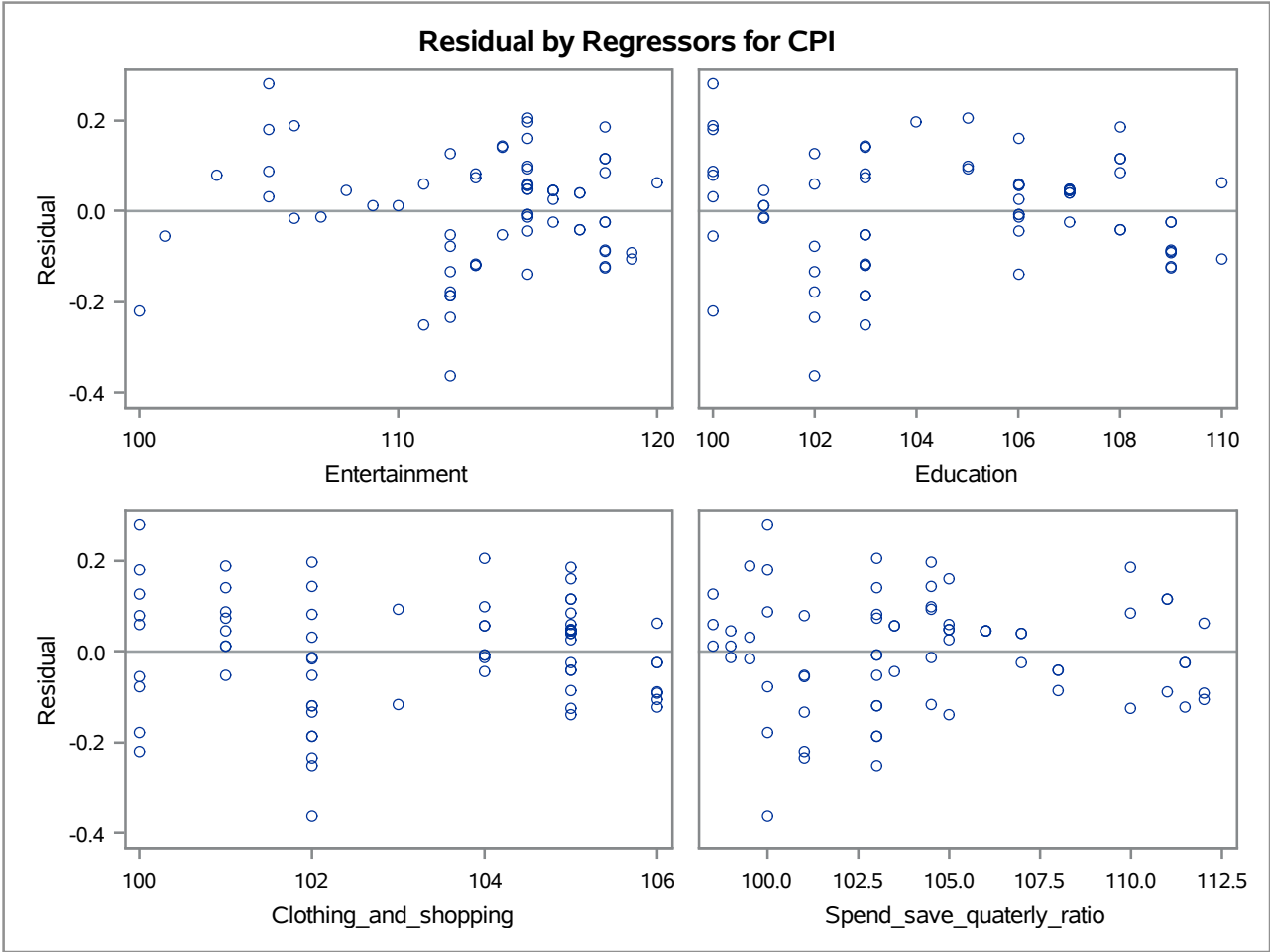
The REG Procedure
Model: MODEL1
Dependent Variable: CPI CPI



Selection criteria: SBC | Details=All shows step-by-step removal log

SECTION 6B: Multivariate Regression — Backward Selection
Starts with all 9 variables, removes the weakest predictor at each step
Response: CPI | Predictors: 9 Consumer Spending Categories

The REG Procedure
Model: MODEL1
Dependent Variable: CPI CPI



SECTION 6C: Multivariate Regression — MaxR Selection
Maximizes R-squared by evaluating the best variable swap at each step
Response: CPI | Predictors: 9 Consumer Spending Categories

The REG Procedure
Model: MODEL1
Dependent Variable: CPI CPI

Number of Observations Read	75
Number of Observations Used	69
Number of Observations with Missing Values	6

Maximum R-Square Improvement: Step 1

Variable Spend_save_quaterly_ratio Entered: R-Square = 0.8661 and C(p) = 101.1809

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	1	18.25606	18.25606	433.27	<.0001
Error	67	2.82307	0.04214		
Corrected Total	68	21.07913			

Variable	Parameter Estimate	Standard Error	Type II SS	F Value	Pr > F
Intercept	91.90309	0.64805	847.39683	20111.3	<.0001
Spend_save_quaterly_ratio	0.12923	0.00621	18.25606	433.27	<.0001

Bounds on condition number: 1, 1

The above model is the best 1-variable model found.

Maximum R-Square Improvement: Step 2

Variable Clothing_and_shopping Entered: R-Square = 0.9050 and C(p) = 54.8481

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	2	19.07714	9.53857	314.46	<.0001
Error	66	2.00199	0.03033		
Corrected Total	68	21.07913			

Variable	Parameter Estimate	Standard Error	Type II SS	F Value	Pr > F
Intercept	85.53920	1.34109	123.40594	4068.34	<.0001
Clothing_and_shopping	0.11089	0.02131	0.82107	27.07	<.0001
Spend_save_quaterly_ratio	0.08058	0.01073	1.70936	56.35	<.0001

MaxR differs from Stepwise: substitutions evaluated before removals

SECTION 6C: Multivariate Regression — MaxR Selection
Maximizes R-squared by evaluating the best variable swap at each step
Response: CPI | Predictors: 9 Consumer Spending Categories

The REG Procedure
Model: MODEL1
Dependent Variable: CPI CPI

Maximum R-Square Improvement: Step 2

Bounds on condition number: 4.1518, 16.607

The above model is the best 2-variable model found.

Maximum R-Square Improvement: Step 3

Variable Entertainment Entered: R-Square = 0.9374 and C(p) = 16.7114

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	3	19.75898	6.58633	324.29	<.0001
Error	65	1.32016	0.02031		
Corrected Total	68	21.07913			

Variable	Parameter Estimate	Standard Error	Type II SS	F Value	Pr > F
Intercept	83.43388	1.15596	105.80571	5209.52	<.0001
Entertainment	-0.04118	0.00711	0.68184	33.57	<.0001
Clothing_and_shopping	0.16070	0.01944	1.38735	68.31	<.0001
Spend_save_quaterly_ratio	0.09622	0.00919	2.22716	109.66	<.0001

Bounds on condition number: 5.161, 39.611

Maximum R-Square Improvement: Step 4

Variable Clothing_and_shopping Removed: R-Square = 0.9396 and C(p) = 13.9862
Variable Education Entered

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	3	19.80527	6.60176	336.86	<.0001
Error	65	1.27386	0.01960		
Corrected Total	68	21.07913			

MaxR differs from Stepwise: substitutions evaluated before removals

SECTION 6C: Multivariate Regression — MaxR Selection
Maximizes R-squared by evaluating the best variable swap at each step
Response: CPI | Predictors: 9 Consumer Spending Categories

The REG Procedure
Model: MODEL1
Dependent Variable: CPI CPI

Maximum R-Square Improvement: Step 4

Variable	Parameter Estimate	Standard Error	Type II SS	F Value	Pr > F
Intercept	87.62491	0.70163	305.66773	15597.0	<.0001
Entertainment	-0.07407	0.00931	1.24195	63.37	<.0001
Education	0.18226	0.02131	1.43365	73.15	<.0001
Spend_save_quaterly_ratio	0.06784	0.01131	0.70560	36.00	<.0001

Bounds on condition number: 14.868, 84.64

The above model is the best 3-variable model found.

Maximum R-Square Improvement: Step 5

Variable Clothing_and_shopping Entered: R-Square = 0.9513 and C(p) = 1.3699

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	4	20.05357	5.01339	312.86	<.0001
Error	64	1.02556	0.01602		
Corrected Total	68	21.07913			

Variable	Parameter Estimate	Standard Error	Type II SS	F Value	Pr > F
Intercept	84.33993	1.04830	103.72301	6472.84	<.0001
Entertainment	-0.06644	0.00863	0.94870	59.20	<.0001
Education	0.11236	0.02620	0.29460	18.38	<.0001
Clothing_and_shopping	0.09246	0.02349	0.24830	15.50	0.0002
Spend_save_quaterly_ratio	0.06974	0.01024	0.74399	46.43	<.0001

Bounds on condition number: 27.497, 202.93

The above model is the best 4-variable model found.

MaxR differs from Stepwise: substitutions evaluated before removals

SECTION 6C: Multivariate Regression — MaxR Selection
Maximizes R-squared by evaluating the best variable swap at each step
Response: CPI | Predictors: 9 Consumer Spending Categories

The REG Procedure
Model: MODEL1
Dependent Variable: CPI CPI

Maximum R-Square Improvement: Step 6

Variable Furniture_Home_Improvement Entered: R-Square = 0.9517 and C(p) = 2.9554

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	5	20.06061	4.01212	248.17	<.0001
Error	63	1.01852	0.01617		
Corrected Total	68	21.07913			

Variable	Parameter Estimate	Standard Error	Type II SS	F Value	Pr > F
Intercept	84.40323	1.05731	103.02395	6372.50	<.0001
Furniture_Home_Improvement	0.00767	0.01162	0.00704	0.44	0.5117
Entertainment	-0.07587	0.01671	0.33320	20.61	<.0001
Education	0.12025	0.02891	0.27973	17.30	<.0001
Clothing_and_shopping	0.08254	0.02797	0.14074	8.71	0.0044
Spend_save_quaterly_ratio	0.07275	0.01125	0.67641	41.84	<.0001

Bounds on condition number: 33.167, 444.45

The above model is the best 5-variable model found.

Maximum R-Square Improvement: Step 7

Variable Travel_including_Leisure Entered: R-Square = 0.9521 and C(p) = 4.3937

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	6	20.07016	3.34503	205.55	<.0001
Error	62	1.00898	0.01627		
Corrected Total	68	21.07913			

Variable	Parameter Estimate	Standard Error	Type II SS	F Value	Pr > F
Intercept	83.31267	1.77582	35.81913	2201.03	<.0001
Furniture_Home_Improvement	0.00942	0.01188	0.01023	0.63	0.4310
Travel_including_Leisure	-0.00655	0.00855	0.00954	0.59	0.4467
Entertainment	-0.06974	0.01858	0.22934	14.09	0.0004

MaxR differs from Stepwise: substitutions evaluated before removals

SECTION 6C: Multivariate Regression — MaxR Selection
Maximizes R-squared by evaluating the best variable swap at each step
Response: CPI | Predictors: 9 Consumer Spending Categories

The REG Procedure
Model: MODEL1
Dependent Variable: CPI CPI

Maximum R-Square Improvement: Step 7

Variable	Parameter Estimate	Standard Error	Type II SS	F Value	Pr > F
Education	0.12287	0.02920	0.28805	17.70	<.0001
Clothing_and_shopping	0.09037	0.02987	0.14894	9.15	0.0036
Spend_save_quaterly_ratio	0.07167	0.01137	0.64629	39.71	<.0001

Bounds on condition number: 33.629, 705.41

The above model is the best 6-variable model found.

Maximum R-Square Improvement: Step 8

Variable Communication Entered: R-Square = 0.9524 and C(p) = 6.0016

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	7	20.07682	2.86812	174.55	<.0001
Error	61	1.00231	0.01643		
Corrected Total	68	21.07913			

Variable	Parameter Estimate	Standard Error	Type II SS	F Value	Pr > F
Intercept	83.23078	1.78902	35.56396	2164.39	<.0001
Furniture_Home_Improvement	0.00901	0.01195	0.00933	0.57	0.4539
Travel_including_Leisure	-0.00739	0.00869	0.01186	0.72	0.3988
Entertainment	-0.06744	0.01901	0.20669	12.58	0.0008
Education	0.11998	0.02969	0.26828	16.33	0.0002
Communication	-0.00094741	0.00149	0.00666	0.41	0.5267
Clothing_and_shopping	0.09286	0.03027	0.15465	9.41	0.0032
Spend_save_quaterly_ratio	0.07274	0.01155	0.65162	39.66	<.0001

Bounds on condition number: 34.431, 850.97

The above model is the best 7-variable model found.

MaxR differs from Stepwise: substitutions evaluated before removals

SECTION 6C: Multivariate Regression — MaxR Selection
Maximizes R-squared by evaluating the best variable swap at each step
Response: CPI | Predictors: 9 Consumer Spending Categories

The REG Procedure
Model: MODEL1
Dependent Variable: CPI CPI

Maximum R-Square Improvement: Step 9

Variable Grocery Entered: R-Square = 0.9525 and C(p) = 8.0002

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	8	20.07684	2.50960	150.23	<.0001
Error	60	1.00229	0.01670		
Corrected Total	68	21.07913			

Variable	Parameter Estimate	Standard Error	Type II SS	F Value	Pr > F
Intercept	83.14480	2.91172	13.62114	815.40	<.0001
Furniture_Home_Improvement	0.00901	0.01205	0.00934	0.56	0.4576
Travel_including_Leisure	-0.00741	0.00878	0.01188	0.71	0.4024
Entertainment	-0.06759	0.01959	0.19883	11.90	0.0010
Grocery	0.00130	0.03461	0.00002364	0.00	0.9701
Education	0.11976	0.03049	0.25767	15.42	0.0002
Communication	-0.00094349	0.00150	0.00657	0.39	0.5328
Clothing_and_shopping	0.09292	0.03056	0.15446	9.25	0.0035
Spend_save_quaterly_ratio	0.07263	0.01205	0.60671	36.32	<.0001

Bounds on condition number: 35.718, 1083.5

The above model is the best 8-variable model found.

Maximum R-Square Improvement: Step 10

Variable Eating_out Entered: R-Square = 0.9525 and C(p) = 10.0000

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	9	20.07684	2.23076	131.31	<.0001
Error	59	1.00229	0.01699		
Corrected Total	68	21.07913			

MaxR differs from Stepwise: substitutions evaluated before removals

SECTION 6C: Multivariate Regression — MaxR Selection
Maximizes R-squared by evaluating the best variable swap at each step
Response: CPI | Predictors: 9 Consumer Spending Categories

The REG Procedure
Model: MODEL1
Dependent Variable: CPI CPI

Maximum R-Square Improvement: Step 10

Variable	Parameter Estimate	Standard Error	Type II SS	F Value	Pr > F
Intercept	83.13330	3.04430	12.66819	745.72	<.0001
Furniture_Home_Improvement	0.00911	0.01382	0.00738	0.43	0.5125
Travel_including_Leisure	-0.00750	0.01098	0.00793	0.47	0.4972
Eating_out	-0.00015179	0.01061	0.00000348	0.00	0.9886
Entertainment	-0.06756	0.01989	0.19602	11.54	0.0012
Grocery	0.00123	0.03527	0.00002060	0.00	0.9723
Education	0.11989	0.03200	0.23849	14.04	0.0004
Communication	-0.00094288	0.00152	0.00656	0.39	0.5367
Clothing_and_shopping	0.09299	0.03123	0.15059	8.86	0.0042
Spend_save_quaterly_ratio	0.07273	0.01410	0.45186	26.60	<.0001

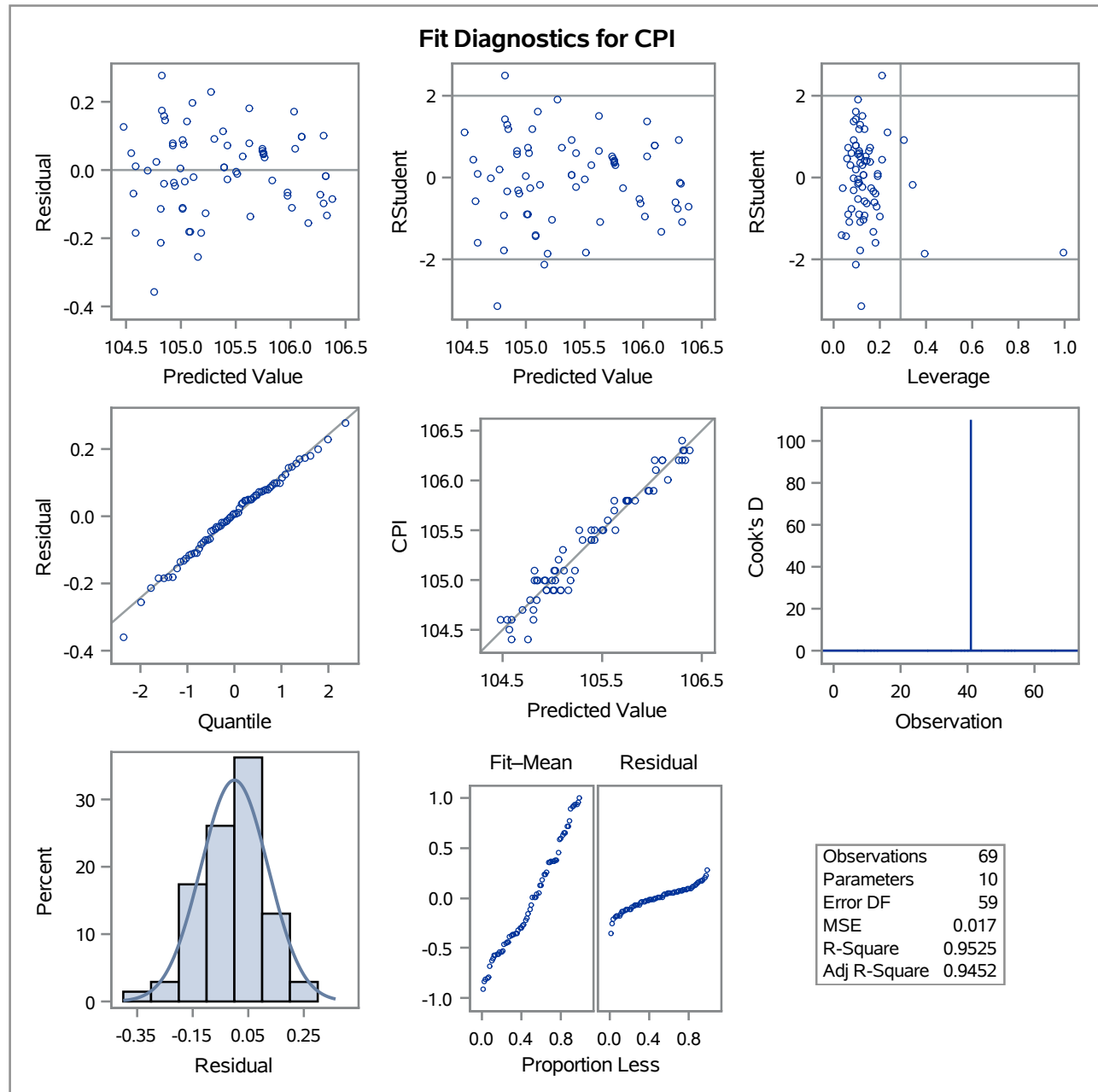
Bounds on condition number: 38.673, 1478

The above model is the best 9-variable model found.

No further improvement in R-Square is possible.

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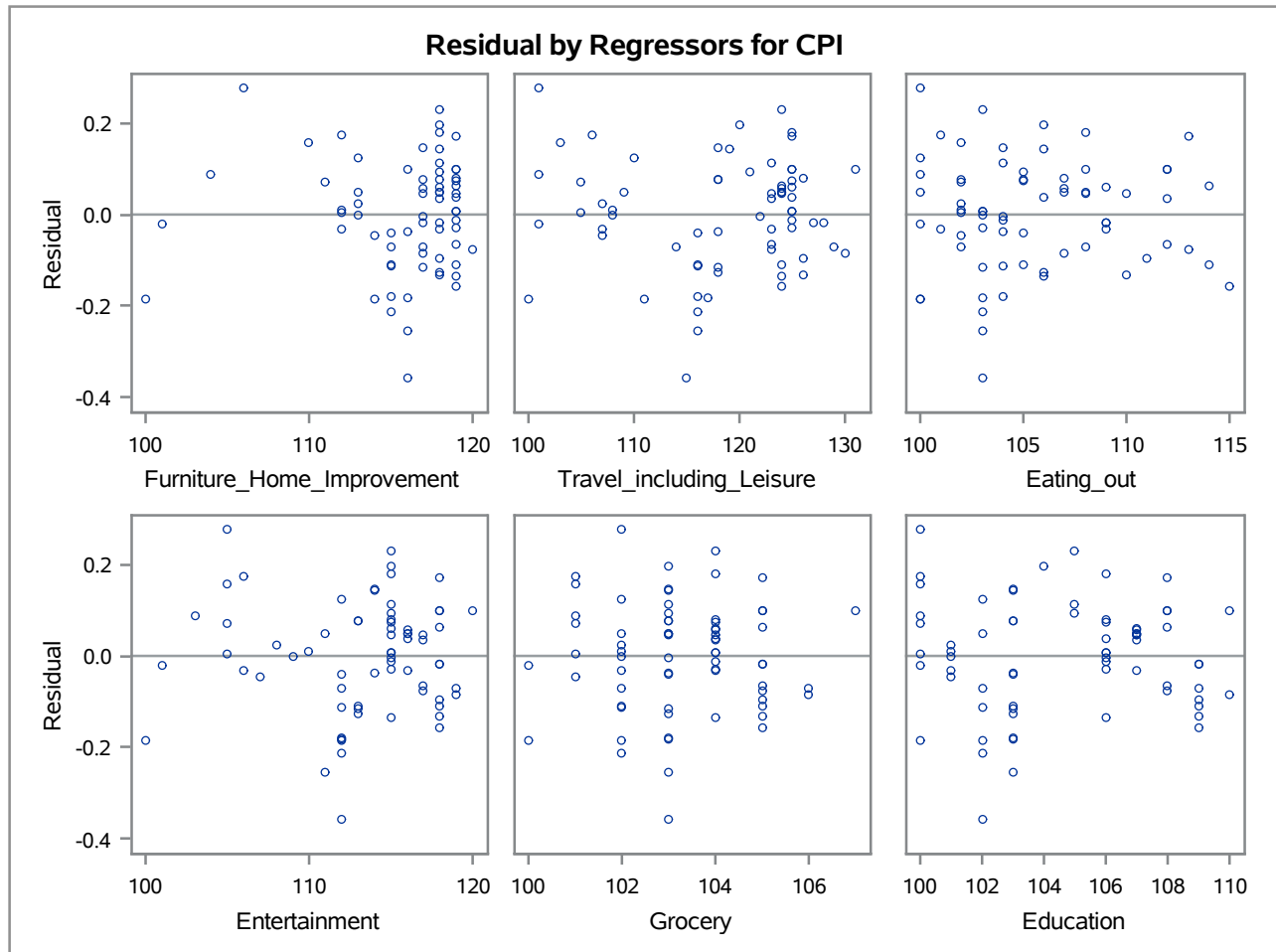
The REG Procedure
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Dependent Variable: CPI CPI



MaxR differs from Stepwise: substitutions evaluated before removals

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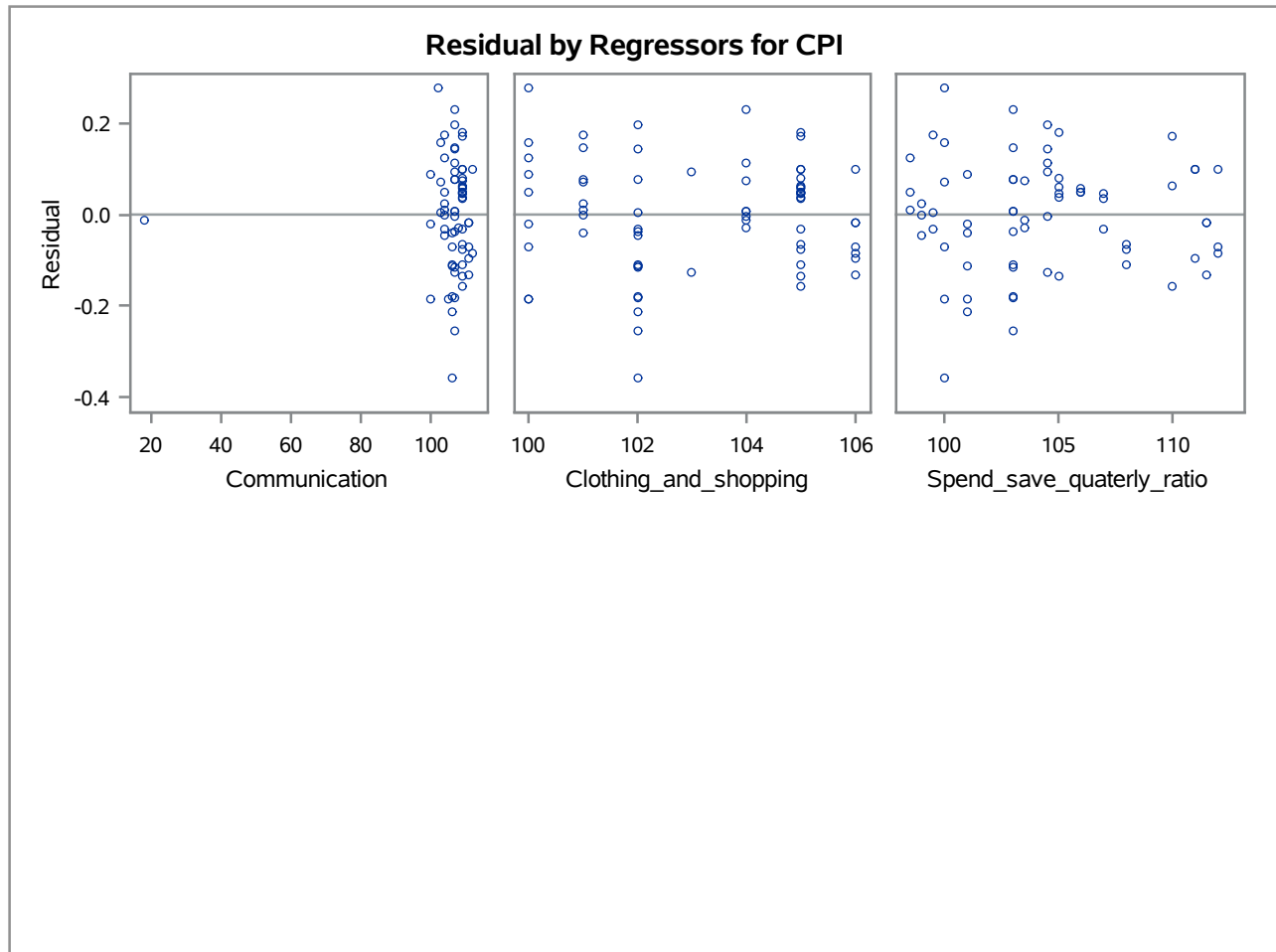
The REG Procedure
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MaxR differs from Stepwise: substitutions evaluated before removals

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MaxR differs from Stepwise: substitutions evaluated before removals